Training: mnist-smoke

Purpose and interpretation

This run fits the pre-specified velocity MSE objective. Completion and held-out error establish execution; generation quality requires its separate evaluation. No checkpoint was selected by best validation or test score.

Recorded evidence


```
{
  "status": "step_limit",
  "dataset": "mnist",
  "family": "onn",
  "seed": 101,
  "step": 100,
  "train_seconds": 33.96222671099531,
  "parameters": 1289488,
  "device": "cuda",
  "initial_validation_mse": 2.111804723739624,
  "final_validation_mse": 2.0086912736296654,
  "images_per_second": 376.88930437108195,
  "source_revisions": [
    "a3dd6228d6e9296a8555102d0b854b53e4a547aa"
  ],
  "peak_memory_bytes": 203779584,
  "checkpoint_sha256":
"8f38d4c79f09a651583cc67b6596ab438a75269ad8fcb6a89db4135257a2a9db",
  "failure": null,
  "id": "mnist-smoke",
  "artifact": "runs/checks/mnist-smoke/training.json"
}
```


Exact configuration


```
{
  "model": {
    "family": "onn",
    "channels": 1,
    "size": 28,
    "width": 128,
    "depth": 4,
    "patch": 4,
    "rotating_dim": 4,
    "inner_steps": 4,
    "step_size": 0.1,
    "ffn_hidden": 0
  },
  "dataset": "mnist",
  "seed": 101,
  "device": "cuda",
  "batch_size": 128,
  "microbatch": 32,
  "learning_rate": 0.0003,
```

```
  "weight_decay": 0.01,
  "clip_grad": 1.0,
  "ema_decay": 0.999,
  "minutes": 3.0,
  "max_steps": 100,
  "validation_size": 5000,
  "data_root": "data",
  "runs_root": "runs",
  "budget_hours": 24.0,
  "log_every": 50,
  "checkpoint_seconds": 300.0,
  "amp": true
}
```


Method choices and learning

The shared optimizer, data policy, architecture, and stopping rule were chosen before final comparison and are described in the technical paper. The 3e-4 learning rate is a conservative AdamW starting point, not a tuned optimum. Microbatches bound laptop memory; four internal updates bound recurrent cost; EMA smooths noisy parameter updates. These choices trade study breadth for a fixed local compute budget. This record supports only the measurements above; causes of quality differences require the trained controls and uncertainty analysis. Full trajectories, populations, RNG states, and logs remain in the referenced artifact bundle. Sources: protocol, TorchCFM, AKOrN.